

國立高雄大學九十五學年度研究所碩士班招生考試試題

科目：統計學

系所：經濟管理研究所全球經貿組



可 使用計算機

考試時間：100 分鐘

本科原始成績：滿分 100 分



否

4. Let $X_1, X_2, \dots, X_n$ be independent, uniformly distributed random variables on the interval $[0, \theta]$. (20%)

(a) Find the maximum likelihood estimator $\hat{\theta}$ of θ .

(b) Is $\hat{\theta}$ an unbiased estimator of θ ?

(c) Is $\hat{\theta}$ a consistent estimator of θ ?

5. For a given experiment, S denotes the sample space and $A_1, A_2, A_3, \dots$ represent possible events. Assume that a number $P(A)$, called the probability of A , satisfies the following three axioms:

(Axiom 1) $P(A) \geq 0$ for every A ;

(Axiom 2) $P(S) = 1$;

(Axiom 3) $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$ if $A_i \cap A_j = \emptyset$ for $i \neq j$.

Consider the experiment of rolling a fair die, where the experimental outcome is defined as the number of dots appearing on the upward face of the die. Show that the probability that one dot appears on the upward face of the die is $1/6$.

(15%)

Area in Upper Tail

t Distribution		
Degrees of Freedom	Upper 5% point	Upper 2.5% point
6	2.447	3.143
7	2.365	2.998
8	2.306	2.896
9	2.262	2.821
10	2.228	2.821
$\vdots$	$\vdots$	$\vdots$
∞	1.645	1.96